



Asset Performance:

Pol 03 Local air quality



Credits



No Minimum Standard

Aim

To reduce asset contribution to local air pollution by using no or low emission heating and hot water systems in the asset.

Question

Do the asset's heating and hot water systems generate local emissions of nitrogen oxides, particulate matter or volatile organic compounds?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	Yes, emissions from combustion appliance(s) exceed the limits in Table 26
1	C.	Yes, emissions from biomass or solid fuel combustion appliance(s) comply with the limits in Table 26
2	D.	Yes, emissions from oil combustion appliance(s) comply with the limits in Table 26
3	E.	Yes, emissions from gas combustion appliance(s) comply with the limits in Table 26
4	F.	No, all heating and hot water is supplied by non-combustion system(s), e.g. powered by electricity

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where the asset is connected to a district heating system that is outside the control of the building owner or manager, this issue can be filtered out of the assessment.	All
2.	Credits are awarded where emissions from the asset's combustion appliance(s) that provide space heating and domestic hot water to the asset	B – E

Criterion	Assessment criteria	Applicable Answer
	do not exceed the limits set in Table 26 in the Checklists and Tables section of this issue.	
3.	The emission limits in Table 26 use the following units of measurement: a) Nitrogen oxides (NO_x) measured in mg/kWh fuel input based on Gross Calorific Value (GCV) for gas or oil appliances. b) Particulate matter and volatile organic compounds (VOC) for all solid fuel or biomass boilers measured in mg/m³ 10% O₂ dry basis. c) Particulate matter and VOC for all solid fuel or biomass local heaters measured in mg/m³ 13% O₂ dry basis. To demonstrate compliance with the emission limits, emission measurement information must be provided by the manufacturer(s) of the appliance(s). Where emission information is not available in these units of measurement, the correct information must be obtained from the appliance manufacturer.	B – E
4.	Where multiple appliances are installed, credits are awarded based on the worst performing appliance.	B – F
5.	Back-up space or water heating appliances can be excluded from assessment. This is on the basis that these appliances will only be used in an emergency so their impact will be limited.	B – F
6.	No credits may be awarded if any combustion appliances are not covered in Table 26, e.g. open fronted or open flue heaters.	B – E
7.	Where a project is connected to a district heating system that is outside the control of the building owner or manager, the system does not need to be assessed. Where the building owner or manager does have control over the system, then it must be assessed.	B – F

Checklists and Tables

Table 26: Emission limit from combustion appliances

Appliance type	Fuel	Nitrogen oxides emissions	Particulate matter emissions	Volatile organic compound emissions
Boiler space heaters or combination heaters or water heaters	Gas	56 mg/kWh		
	Oil	120 mg/kWh		
	Biomass	200 mg/m ³	40 mg/m ³	20 mg/m ³
	Solid fuel	350 mg/m ³	40 mg/m ³	20 mg/m ³

Appliance type	Fuel	Nitrogen oxides emissions	Particulate matter emissions	Volatile organic compound emissions
Cogeneration space heaters using external combustion	Gas	70 mg/kWh		
	Oil	120 mg/kWh		
Cogeneration space heaters using internal combustion engine	Gas	240 mg/kWh		
	Oil	420 mg/kWh		
Heat pump space heaters or combination heaters or water heaters using external combustion	Gas	70 mg/kWh		
	Oil	120 mg/kWh		
Heat pump space heaters or combination heaters or water heaters using internal combustion engine	Gas	240 mg/kWh		
	Oil	420 mg/kWh		
Local space heaters	Gas	130 mg/kWh		
	Oil	130 mg/kWh		
Closed fronted local space heaters	Wood pellets	200 mg/m ³	20 mg/m ³	60 mg/m ³
	Biomass	200 mg/m ³	40 mg/m ³	120 mg/m ³
	Solid fuel	300 mg/m ³	40 mg/m ³	120 mg/m ³

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Copy of manufacturer's details for installed appliance(s) and their emissions levels.
All	Photographic evidence of heating and hot water system(s).

Additional information

Emission limits

The emission limits and measurement units are based on the requirements of the European Union Ecodesign Directive (2009/125/EC) and its associated regulations for energy-related products. These set performance requirements for combustion-powered heating systems and requires manufacturers to publish the NO_x, particulate matter and VOC emission levels for their product.